

Mini-Project: TATA NEU Sales Analysis

Objective: The purpose of this project is to analyse sales data from TATA NEU using SQL. You are provided with two datasets, TATA_TB1 and TATA_TB2, which contain detailed sales transactions. Your task is to write SQL queries to extract meaningful insights, perform data aggregation, and generate reports based on the datasets.

Problem Statement:

TATA NEU operates across multiple regions, countries, product categories, and customer segments, generating a high volume of transactional data daily. While this data is rich in detail, it is stored across multiple tables and lacks an integrated analytical view. As a result, business stakeholders face difficulty in clearly understanding **overall sales performance, profitability drivers, customer purchasing behaviour, and regional contribution**.

Without a consolidated analysis, identifying high-performing products, profitable customers, key regions, and time-based trends becomes challenging, limiting the organization's ability to make informed, data-driven strategic decisions.

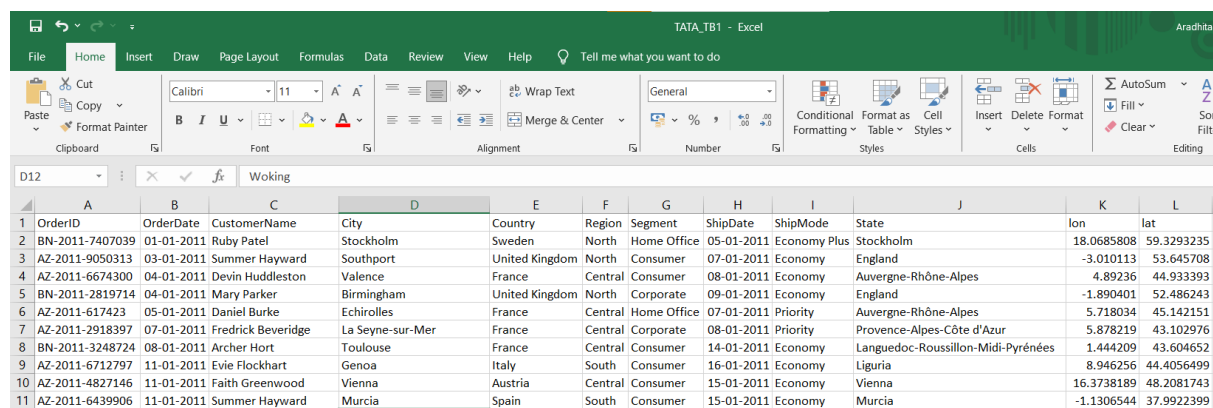
Analysis & Solution Statement:

To address this challenge, sales data from **TATA_TB1** (order, customer, geographic, and shipping information) and **TATA_TB2** (product, category, sales, profit, and quantity details) was integrated and analysed using SQL. The analysis focused on transforming raw transactional data into meaningful business insights by performing data aggregation, filtering, ranking, and segmentation.

Key performance indicators such as **total sales, total profit, order quantity, customer count, product performance, and geographic contribution** were evaluated across multiple dimensions including **time, region, country, category, sub-category, shipping mode, and customer segment**. Advanced SQL techniques such as joins, window functions and stored procedures were implemented to ensure efficient, repeatable, and scalable reporting.

This structured analysis provides TATA NEU leadership with a **comprehensive and actionable view of business performance**, enabling better decision-making related to customer strategy, product optimization, regional focus, and long-term growth planning.

TATA_TB1



OrderID	OrderDate	CustomerName	City	Country	Region	Segment	ShipDate	ShipMode	State	lon	lat
BN-2011-7407039	01-01-2011	Ruby Patel	Stockholm	Sweden	North	Home Office	05-01-2011	Economy Plus	Stockholm	18.0685808	59.3293235
AZ-2011-9050313	03-01-2011	Summer Hayward	Southport	United Kingdom	North	Consumer	07-01-2011	Economy	England	-3.010113	53.645708
AZ-2011-6674300	04-01-2011	Devin Huddleston	Valence	France	Central	Consumer	08-01-2011	Economy	Auvergne-Rhône-Alpes	4.89236	44.933393
BN-2011-2819714	04-01-2011	Mary Parker	Birmingham	United Kingdom	North	Corporate	09-01-2011	Economy	England	-1.890401	52.486243
AZ-2011-617423	05-01-2011	Daniel Burke	Echirolles	France	Central	Home Office	07-01-2011	Priority	Auvergne-Rhône-Alpes	5.718034	45.142151
AZ-2011-2918397	07-01-2011	Fredrick Beveridge	La Seyne-sur-Mer	France	Central	Corporate	08-01-2011	Priority	Provence-Alpes-Côte d'Azur	5.878219	43.102976
BN-2011-3248724	08-01-2011	Archer Hort	Toulouse	France	Central	Consumer	14-01-2011	Economy	Languedoc-Roussillon-Midi-Pyrénées	1.444209	43.604652
AZ-2011-6712797	11-01-2011	Evie Flockhart	Genoa	Italy	South	Consumer	16-01-2011	Economy	Liguria	8.946256	44.4056499
AZ-2011-4827146	11-01-2011	Faith Greenwood	Vienna	Austria	Central	Consumer	15-01-2011	Economy	Vienna	16.3738189	48.2081743
AZ-2011-6439906	11-01-2011	Summer Hayward	Murcia	Spain	South	Consumer	15-01-2011	Economy	Murcia	-1.1306544	37.9922399

TATA_TB2

TATA_TB2 - Excel

	A	B	C	D	E	F	G	H
1	OrderID	ProductName	Discount	Sales	Profit	OrderQuantity	Category	SubCategory
2	BN-2011-7407039	Enermax Note Cards, Premium	0.5	45	-26	3	Office Supplies	Paper
3	AZ-2011-9050313	Dania Corner Shelving, Traditional	0	854	290	7	Furniture	Bookcases
4	AZ-2011-6674300	Binney & Smith Sketch Pad, Easy-Erase	0	140	21	3	Office Supplies	Art
5	BN-2011-2819714	Boston Markers, Easy-Erase	0.5	27	-22	2	Office Supplies	Art
6	BN-2011-2819714	Eldon Folders, Single Width	0.5	17	-1	2	Office Supplies	Storage
7	AZ-2011-617423	Binney & Smith Pencil Sharpener, Water Color	0	90	21	3	Office Supplies	Art
8	AZ-2011-617423	Sanford Canvas, Fluorescent	0	207	77	4	Office Supplies	Art

SQL Query Execution

```

6      #1) Write a query to calculate the total records in both tables?
7
8      •   SELECT (SELECT COUNT(*) FROM tata_tb1) AS tb1_records,
9          (SELECT COUNT(*) FROM tata_tb2) AS tb2_records;

```

	tb1_records	tb2_records
▶	4117	7519

```

11     #2) Write a query to calculate the total unique count of customers?
12
13     •   select count(distinct customername) as unique_customers from tata_tb1;
14

```

	unique_customers
▶	792

```

15     #3) Write a query to fetch the oldest order date and latest order date?
16
17     •   select min(orderdate) as oldest_order_date ,max(orderdate) as latest_order_date from tata_tb1;

```

	oldest_order_date	latest_order_date
▶	2011-01-01	2014-12-31

19 #4) Write query to get years in table?

20

21 • `select distinct year(orderdate) as years from tata_tb1;`

22

<
Result Grid
Filter Rows:
Export:  Wrap Cell Content: 
years
2011
2012
2013
2014

23 #5) Write a query to get the no. of regions and display the region names?

24

25 • `SELECT COUNT(DISTINCT region) AS number_of_regions`

26 `FROM tata_tb1;`

27 • `SELECT DISTINCT region AS region_name`

28 `FROM tata_tb1;`

29

Result Grid
Filter Rows:
region_name
North
Central
South

Result Grid
number_of_regions
3

30 #6) Write a query to get the no. of countries and display the country names?

31

32 • `SELECT COUNT(DISTINCT country) AS number_of_countries`

33 `FROM tata_tb1;`

34 • `SELECT DISTINCT country AS country_name`

35 `FROM tata_tb1;`

Result Grid		Result Grid	
	number_of_countries		country_name
▶	15	▶	Sweden
			United Kingdom
			France
			Italy
			Austria
			Spain
			Germany
			Netherlands
			Denmark
			Belgium
			Norway
			Portugal
			Switzerland
			Ireland
			Finland

```

37 #7) Write a query to get the no. of states and display the state names?
38
39 • SELECT COUNT(DISTINCT state) AS number_of_states
40 FROM tata_tbl;
41 • SELECT DISTINCT state AS state_name
42 FROM tata_tbl;

```

Result Grid		Result Grid	
	number_of_states		state_name
▶	127	▶	Stockholm
			England
			Auvergne-Rhône-Alpes
			Provence-Alpes-Côte d'Azur
			Languedoc-Roussillon-Midi-Pyrénées
			Liguria
			Vienna
			Murcia
			Lower Saxony
			South Holland
			Västra Götaland
			Hovedstaden

```

44      #8) Write a query to get the no. of cities and display the city names?
45
46 •    SELECT COUNT(DISTINCT city) AS number_of_cities
47      FROM tata_tbl;
48 •    SELECT DISTINCT city AS city_name
49      FROM tata_tbl;

```

Result Grid		Result Grid	
	number_of_cities		city_name
▶	999	▶	Stockholm
			Southport
			Valence
			Birmingham
			Echirolles
			La Seyne-sur-Mer
			Toulouse
			Genoa
			Vienna
			Murcia
			Woking
			Lohne

```

51      #9) Write a query to calculate the total count of customers?
52
53 •    SELECT COUNT(distinct CustomerName) AS total_customers FROM tata_tbl;
54

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_customers			
▶	792			

55 #10) Write a query to calculate the total count of products?

56

57 • `SELECT COUNT(distinct ProductName) AS total_products FROM tata_tb2;`

<	
Result Grid	Filter Rows: Export: Wrap Cell Content:
	total_products
▶	1743

59 #11) Write a query to calculate total sales, total profit and total order quantity?

60

61 • `SELECT SUM(Sales) AS total_sales, SUM(Profit) AS total_profit, SUM(OrderQuantity) AS total_order_qty`

62 `FROM TATA_TB2;`

63

<	
Result Grid	Filter Rows: Export: Wrap Cell Content:
	total_sales total_profit total_order_qty
▶	1426144 151633 27307

64 #12) Write a query to calculate the total sales amount for each category. Display the

65 #category, total sales, and total order qty and order by total sales from highest to lowest?

66

67 • `select category,SUM(sales) as total_sales,SUM(OrderQuantity) as total_order_qty`

68 `from tata_tb2`

69 `group by category`

70 `order by total_sales desc;`

<	
Result Grid	Filter Rows: Export: Wrap Cell Content:
	category total_sales total_order_qty
▶	Office Supplies 613522 19230
	Technology 473748 4394
	Furniture 338874 3683

72 #13) Write a query to calculate the total profit amount for each category. Display the

73 #category, total profit, and total order qty and order by total profit from highest to lowest?

74

75 • `select category,SUM(Profit) as total_profit,SUM(OrderQuantity) as total_order_qty`

76 `from tata_tb2`

77 `group by category`

78 `order by total_profit desc;`

<	
Result Grid	Filter Rows: Export: Wrap Cell Content:
	category total_profit total_order_qty
▶	Office Supplies 83064 19230
	Technology 39742 4394
	Furniture 28827 3683


```

80 #14) Write a query to fetch the subcategories where total sales are greater than 100000?
81
82 • select subcategory,sum(sales) as total_sales from tata_tb2
83     group by subcategory
84     having total_sales>100000;

```

	subcategory	total_sales
▶	Bookcases	141434
	Art	127184
	Storage	222403
	Chairs	107161
	Machines	109346
	Phones	135454
	Copiers	140125

```

86 #15) Write a query to fetch the products where total profit is greater than 1000?
87
88 • select productname,sum(profit) as total_profit from tata_tb2
89     group by productname
90     having total_profit>1000;

```

	productname	total_profit
▶	Dania Corner Shelving, Traditional	1111
	Rogers Lockers, Single Width	1172
	Office Star Rocking Chair, Red	1105
	Eldon File Cart, Single Width	2098
	Tenex File Cart, Single Width	1028
	Cardinal Binding Machine, Clear	1195

```

92 #16) Write a query to get the total sales and total profit for Office Supplies category?
93
94 • SELECT SUM(Sales) AS total_sales, SUM(Profit) AS total_profit
95     FROM TATA_TB2
96     where category='Office Supplies';

```

	total_sales	total_profit
▶	613522	83064

98 #17) Write a query to get the total sales and total profit for Furniture category and the Tables
99 #and Bookcases sub-categories ?

100

```
101 • SELECT SubCategory, SUM(Sales) AS total_sales, SUM(Profit) AS total_profit
102 FROM TATA_TB2
103 where category='Furniture' and subcategory in("Tables","Bookcases")
104 GROUP BY SubCategory;
```

105

<

Result Grid Filter Rows: Export: Wrap Cell Content:

	SubCategory	total_sales	total_profit
▶	Bookcases	141434	20794
	Tables	23062	-7372

106 #18) Write a query to get the total sales and total profit for Technology category and the
107 #Accessories, Copiers, Phones sub-categories ?

108

```
109 • SELECT SubCategory, SUM(Sales) AS total_sales, SUM(Profit) AS total_profit
110 FROM TATA_TB2
111 where category='Technology' and subcategory in('Accessories', 'Copiers', 'Phones')
112 GROUP BY SubCategory;
```

113

<

Result Grid Filter Rows: Export: Wrap Cell Content:

	SubCategory	total_sales	total_profit
▶	Phones	135454	5066
	Copiers	140125	16099
	Accessories	88823	14826

114 #19) Write a query to get total sales and total profit by Region, Segment?

115

```
116 • SELECT t1.Region, t1.Segment, SUM(t2.Sales) AS total_sales, SUM(t2.Profit) AS total_profit
117 FROM TATA_TB1 t1
118 JOIN TATA_TB2 t2 ON t1.OrderID = t2.OrderID
119 GROUP BY t1.Region, t1.Segment;
```

120

<

Result Grid Filter Rows: Export: Wrap Cell Content:

	Region	Segment	total_sales	total_profit
▶	North	Home Office	48652	4605
	North	Consumer	154017	14014
	Central	Consumer	429226	47823
	North	Corporate	101750	11524
	Central	Home Office	138635	18455
	Central	Corporate	231489	30111
	South	Consumer	167612	13192
	South	Corporate	104483	5158
	South	Home Office	50280	6751


```

121 #20) Write a query to get total sales and total profit by Country, State and city?
122
123 • SELECT t1.Country,t1.state,t1.city, SUM(t2.Sales) AS total_sales, SUM(t2.Profit) AS total_profit
124 FROM TATA_TB1 t1 join TATA_TB2 t2 on t1.orderid=t2.orderid
125 group by t1.Country,t1.state,t1.city;

```

126

<

Result Grid Filter Rows: Export: Wrap Cell Content:

	Country	state	city	total_sales	total_profit
▶	Sweden	Stockholm	Stockholm	10768	-5787
	United Kingdom	England	Southport	1010	268
	France	Auvergne-Rhône-Alpes	Valence	983	165
	United Kingdom	England	Birmingham	7944	-4490
	France	Auvergne-Rhône-Alpes	Echirolles	1259	177
	France	Provence-Alpes-Côte d'Azur	La Seyne-sur-Mer	1897	419
	France	Languedoc-Roussillon-Midi-Pyrénées	Toulouse	7949	-6598
	United Kingdom	England	Woking	198	47
	Spain	Murcia	Murcia	4697	546
	Austria	Vienna	Vienna	33202	8748
	Italy	Liguria	Genoa	2634	231
	United Kingdom	England	Sheffield	3478	-1690
	United Kingdom	England	Leicester	1249	362
	Germany	Lower Saxony	Lohne	1740	331
	Netherlands	South Holland	Dordrecht	872	-508

```

127 #21) Write a query to get total sales and total orderqty by CustomerName sort it by totalsales
128 #from highest to lowest?

```

129

```

130 • select t1.customername, SUM(t2.sales) as total_sales, SUM(t2.orderQuantity) as total_orderqty
131 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
132 group by t1.customername
133 order by total_sales desc;

```

134

<

Result Grid Filter Rows: Export: Wrap Cell Content:

	customername	total_sales	total_orderqty
▶	Joel Peters	8092	126
	Angie Massengill	6973	84
	Jose Gambino	6824	124
	Hayden Perkins	6788	77
	Elijah Sodeman	6469	96
	Harry Josephson	6160	82
	Lilly Le Grand	5890	75
	Isaac David	5724	94
	Michael Watkins	5669	94
	Alexis Code	5663	67
	Alexandra Macd...	5496	99
	Mark Washington	5478	90

```

135 #22) Identify the top 5 products with the highest sales (by sales amount). Show the product
136 #name, total sales, and total qty?
137
138 • select productname, SUM(sales) as total_sales, SUM(orderquantity) as total_qty
139 from tata_tb2
140 group by productname
141 order by total_sales desc limit 5;
142

```

Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
Fetch rows:			
	productname	total_sales	total_qty
▶	Rogers File Cart, Single Width	10687	91
	Eldon File Cart, Single Width	10206	96
	Tenex File Cart, Single Width	7484	67
	Eldon Lockers, Industrial	7183	42
	Eldon Lockers, Blue	5976	42

```

143 #23) Write a query to get total sales by City having sales greater than 2000?
144
145 • select t1.city, SUM(t2.sales) as total_sales
146 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
147 group by t1.city
148 having total_sales > 2000;

```

Result Grid		
Filter Rows:		
Export:		
Wrap Cell Content:		
	city	total_sales
▶	Stockholm	10768
	Birmingham	7944
	Toulouse	7949
	Murcia	4697
	Vienna	33202
	Genoa	2634
	Sheffield	3478

```

150 #24) Write a query to get total sales by CustomerName having sales greater than 3000?
151
152 • select t1.customername,SUM(t2.sales) as total_sales
153 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
154 group by t1.customername
155 having total_sales>3000;
156

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	customername	total_sales
▶	Summer Hayward	5017
	Archer Hort	4642
	Evie Flockhart	3503
	Hershel Snyder	3477
	Ella Troy	4272
	Everett Dunbar	4513
	Melissa Bean	3108
	Lori Miller	3659

```

157 #25) Write a query to get total sales and total profit by shipmode?
158
159 • select t1.shipmode,SUM(t2.sales) as total_sales,SUM(t2.profit) as total_profit
160 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
161 group by t1.shipmode;
162

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	shipmode	total_sales	total_profit
▶	Economy Plus	271618	27326
	Economy	878339	97764
	Priority	206664	15869
	Immediate	69523	10674

```

163 #26) Write a query to get total sales for North and central region?
164
165 • select t1.region,SUM(t2.sales) as total_sales
166 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
167 where region in('North','central')
168 group by t1.region;
169

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	region	total_sales
▶	North	304419
	Central	799350

```

172 • select t1.country, SUM(t2.sales) as total_sales, SUM(t2.profit) as total_profit
173 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
174 where country in('Italy', 'Spain')
175 group by t1.country;
176

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	country	total_sales	total_profit
▶	Spain	145591	26430
	Italy	167678	3640

```

177 #28) Write a query to get the total sales and total profit for each year?
178
179 • select year(t1.orderdate) as yr, SUM(t2.sales) as total_sales, SUM(t2.profit) as total_profit
180 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid = t2.orderid
181 group by yr;
182

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	yr	total_sales	total_profit
▶	2011	264916	27330
	2012	323741	36930
	2013	365676	36942
	2014	471811	50431

```

183 #29) Find the top 10 customers who spent the most across all transactions. Display the
184 #customer name, total amount spent, and number of orders placed?
185
186 • select t1.customername, SUM(t2.sales) as total_amt_spent, count(distinct t1.orderid)
187 from tata_tb1 t1 join tata_tb2 t2 on t1.orderid=t2.orderid
188 group by t1.customername
189 order by total_amt_spent desc
190 limit 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#) | Fetch rows: [↕](#)

	customername	total_amt_spent	count(distinct t1.orderid)
▶	Joel Peters	8092	6
	Angie Massengill	6973	10
	Jose Gambino	6824	13
	Hayden Perkins	6788	10
	Elijah Sodeman	6469	10
	Harry Josephson	6160	9
	Lilly Le Grand	5890	11
	Isaac David	5724	10
	Michael Watkins	5669	10
	Alexis Code	5663	8

192 #30) Write a query to find which products are most preferred by customers based on the
 193 #total sales. Display customer name, favorite product9 (top 3 products per each customer), and total sales on that product?
 194
 195 •

```
select * from(
  196     select t1.customername,t2.productname, SUM(t2.sales) as total_sales,
  197     dense_rank() over(partition by customername order by SUM(sales) desc) as rnk
  198     from tata_tb1 t1 join tata_tb2 t2 on t1.orderid=t2.orderid
  199     group by t1.customername,t2.productname)x
  200     WHERE rnk <= 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customername	productname	total_sales	rnk
▶	Aaron Bootman	HP Personal Copier, Digital	632	1
	Aaron Bootman	Belkin Keyboard, Ergonomic	245	2
	Aaron Bootman	Konica Receipt Printer, Durable	245	2
	Aaron Bootman	Tenex File Cart, Blue	240	3
	Aaron Cunningham	Eldon File Cart, Single Width	809	1
	Aaron Cunningham	Hon Rocking Chair, Red	392	2

202 #31) Write a query to get 7th rank customer name based on total sales? Display customer
 203 #name, sales amount and rank.
 204
 205 •

```
select * from(
  206     select t1.customername,SUM(t2.sales) as total_sales,
  207     dense_rank() over(order by SUM(sales) desc) as rnk
  208     from tata_tb1 t1 join tata_tb2 t2 on t1.orderid=t2.orderid
  209     group by t1.customername)x
  210     WHERE rnk = 7;
```

211
 212
 213
 214
 215
 216
 217
 218
 219

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customername	total_sales	rnk
▶	Lilly Le Grand	5890	7

212 #32) Write query to get total sales and total profit between year 2011 and 2013?
 213
 214 •

```
SELECT YEAR(t1.OrderDate) as yr,SUM(t2.Sales) as total_sales, SUM(t2.Profit) as total_profit
  215 FROM TATA_TB1 t1
  216 JOIN TATA_TB2 t2 ON t1.OrderID = t2.OrderID
  217 WHERE YEAR(OrderDate) BETWEEN 2011 AND 2013
  218 group by yr ;
```

219

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	yr	total_sales	total_profit
▶	2011	264916	27330
	2012	323741	36930
	2013	365676	36942

```

220 #33) Write a query to get total sales, total profit and total order qty by country, state, category
221 #and sub-category?
222
223 • select t1.country,t1.state, t2.category,t2.subcategory,SUM(t2.sales),SUM(t2.profit), SUM(t2.orderQuantity)
224 FROM TATA_TB1 t1
225 JOIN TATA_TB2 t2 ON t1.OrderID = t2.OrderID
226 group by t1.country,t1.state, t2.category,t2.subcategory
227 order by country;
228

```

country	state	category	subcategory	SUM(t2.sales)	SUM(t2.profit)	SUM(t2.orderQuantity)
Austria	Carinthia	Furniture	Bookcases	448	81	3
Austria	Carinthia	Furniture	Furnishings	40	8	2
Austria	Carinthia	Office Supplies	Art	412	70	9
Austria	Carinthia	Office Supplies	Binders	59	21	2
Austria	Carinthia	Office Supplies	Envelopes	298	40	10
Austria	Carinthia	Office Supplies	Fasteners	79	39	4
Austria	Carinthia	Office Supplies	Paper	185	74	7
Austria	Carinthia	Office Supplies	Storage	434	116	8
Austria	Carinthia	Office Supplies	Supplies	102	35	6
Austria	Salzburg	Office Supplies	Art	107	9	2
Austria	Salzburg	Office Supplies	Labels	29	10	3

```

229 #34) Write stored procedure to get top 10 customers based on total sales?
230
231 delimiter %%
232 • create procedure top10customers()
233 Begin
234     select t1.customername,SUM(t2.sales) as total_sales
235     FROM TATA_TB1 t1
236     JOIN TATA_TB2 t2 ON t1.OrderID = t2.OrderID
237     group by t1.customername
238     order by total_sales desc limit 10;
239 end ;
240 %%
241 call top10customers();

```

customername	total_sales
Joel Peters	8092
Angie Massengill	6973
Jose Gambino	6824
Hayden Perkins	6788
Elijah Sodeman	6469
Harry Josephson	6160
Lilly Le Grand	5890
Isaac David	5724
Michael Watkins	5669
Alexis Code	5663


```

247      #36) Create a stored procedure to get region sales?
248
249      delimiter $$
250 •    create procedure regionsales()
251      Begin
252      select t1.region,SUM(t2.sales) as total_sales
253      FROM TATA_TB1 t1
254      JOIN TATA_TB2 t2 ON t1.OrderID = t2.OrderID
255      group by t1.region
256      order by total_sales desc;
257      end ;
258      $$
259 •    call regionsales();

```

<		
Result Grid	Filter Rows:	Export: Wrap Cell Content:
	region	total_sales
▶	Central	799350
	South	322375
	North	304419

Reports Generated Based on the Datasets

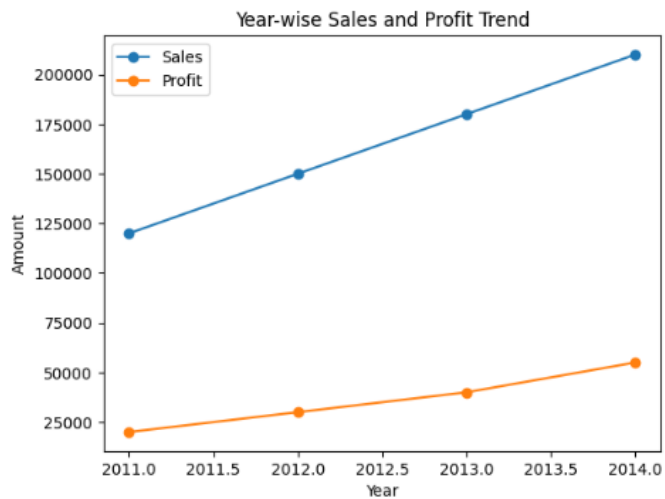
Using data from **TATA_TB1** and **TATA_TB2**, the following analytical reports were generated to provide a comprehensive view of TATA NEU's sales performance:

1. Sales & Profit Summary Report

Provides an overall snapshot of:

- Total sales, total profit, and total order quantity
- Year-wise sales and profit trends
- Sales and profit performance between specific time periods

Purpose: To track overall business health and growth trends.

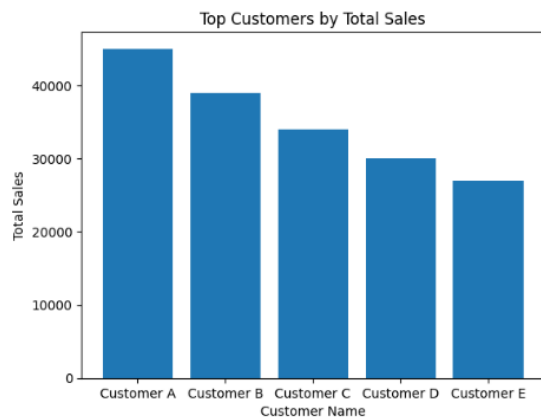


2. Customer Performance Report

Analyzes customer behavior and contribution by:

- Total and unique customer count
- Top customers by total sales and number of orders
- Customer ranking based on sales
- High-value customers contributing the most revenue

Purpose: To identify key customers and support customer retention and targeting strategies.



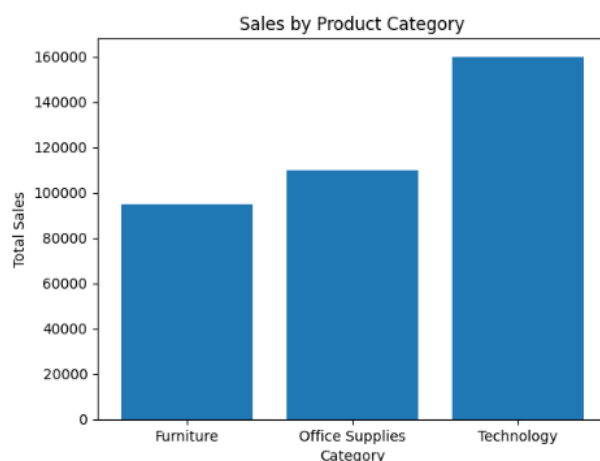
CustomerName	Total Sales
Ruby Patel	45,230
Summer Hayward	38,900
Devin Huddleston	34,120
Mary Parker	29,980
Daniel Burke	26,750

3. Product & Category Performance Report

Evaluates product-level performance including:

- Total number of products sold
- Sales and profit by category and sub-category
- Top-performing and high-profit products
- Customer product preferences (top products per customer)

Purpose: To optimize product portfolio and focus on high-performing categories.



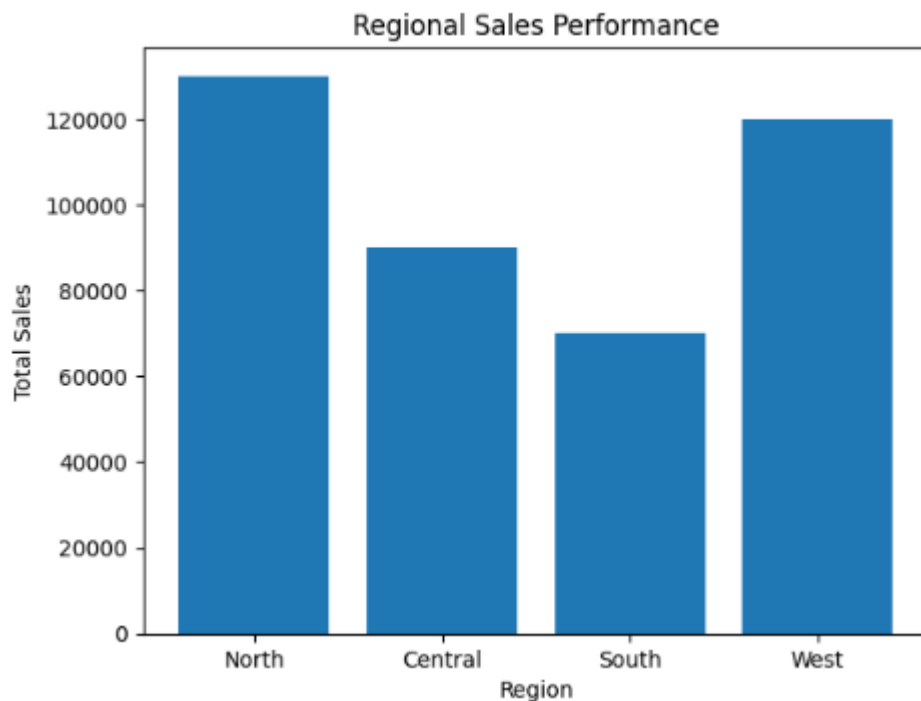
4. Geographic Sales Report

Breaks down performance across locations:

- Sales and profit by region, country, state, and city

- Regional comparisons (e.g., North vs Central)
- Country-specific performance analysis

Purpose: To identify high-performing and underperforming geographic markets.

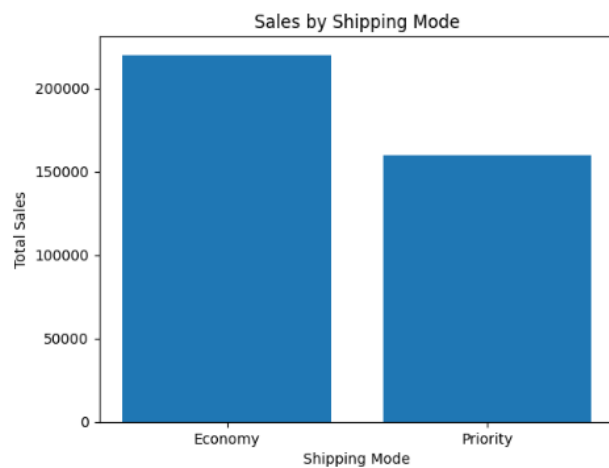


5. Operational & Shipping Report

Assesses operational efficiency by:

- Sales and profit by shipping mode
- Order quantity distribution across shipping methods

Purpose: To support logistics optimization and cost-efficiency decisions.

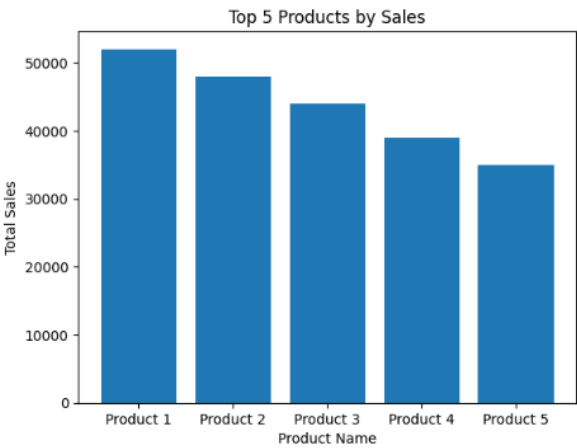


6. Advanced Analytical & Executive Reports

Includes:

- Top 5 products by sales
- Top 10 customers by total spend
- Sales thresholds analysis (cities and customers exceeding target sales)
- Consolidated views and stored procedures for repeatable reporting

Purpose: To enable fast, consistent, and scalable reporting for leadership and analytics teams.



ProductName	Total Sales
Canon imageCLASS Copier	52,100
Cisco Smart Phone	48,300
HP LaserJet Printer	44,200
Logitech Wireless Keyboard	39,100
Samsung Monitor	35,000

Final Outcome

These reports collectively transform raw transactional data into **actionable business insights**, enabling TATA NEU stakeholders to understand performance drivers, identify growth opportunities, and make informed strategic decisions across customers, products, regions, and time.